

Sikao Guo, Ph.D.

RESEARCH SOFTWARE ENGINEER | C++ / PYTHON | SCIENTIFIC COMPUTING

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U.S. Permanent Resident | No sponsorship required

Research software engineer with 6+ years building C++/Python tools for molecular modeling and high-performance computing. Delivered MPI simulation with approximately 90x speedup on 96 CPUs and nine cross-platform applications, from algorithm development and validation through deployment.

TECHNICAL SKILLS

Languages & HPC: C++, Python, MPI, Bash, Linux, CMake, distributed-memory computing

Engineering: API design, JSON Schema, pytest, GitHub Actions, GitLab CI/CD, Docker, Conda/Pixi, PyPI

Interfaces & web: FastAPI, PyQt6, Tkinter, Panel/Bokeh, Nginx, systemd

Scientific libraries: NumPy, SciPy, Biopython, Eigen, CGAL, OpenMM, GROMACS

PROFESSIONAL EXPERIENCE

Inria

Research Engineer

Jul 2025 - Present

Sophia Antipolis, France

- Architected AutoCLIP, a Python framework that generates PyMOL, VMD, and web interfaces from one application specification; deployed **nine tools** using reusable platform adapters.
- Built cross-platform testing and CI/CD for Linux and Intel/Apple Silicon macOS, plus reproducible Pixi/Conda packaging and authenticated web deployment.
- Turned a loop-sampling prototype into a modular C++/Python platform supporting flexible endpoints, multiple loops, rigid secondary-structure motion, and intrinsically disordered regions.
- Implemented spatial-hash collision detection and dense/sparse memory backends; resolved stability and zero-output failures and integrated AttnPacker, OpenMM, and PhiSiCal validation.

Johns Hopkins University

Assistant Research Scientist (2025); Postdoctoral Fellow (2020-2024)

Jan 2020 - Jun 2025

Baltimore, MD

- Led C++/MPI development of NERDSS using spatial domain decomposition and distributed communication; achieved near-linear scaling and approximately **90x speedup on 96 CPUs** for 20,000-particle workloads.
- Verified distributed reactions, assembly, dissociation, and restart against analytical theory and serial results across **seven benchmark models** spanning diffusion, membrane reactions, and clathrin assembly.
- Built ioNERDSS to convert PDB/mmCIF structures into simulation models using KD-tree interface detection, reaction generation, and sequence alignment; benchmarked **44,000+ structures** and 3-720-subunit assemblies.
- Released simulation, restart, analysis, and visualization workflows on PyPI, with PyQt6 interfaces, Docker/Conda environments, automated tests, and Sphinx documentation.

SELECTED OPEN-SOURCE SOFTWARE

AutoCLIP | scientific GUI generation and deployment | [Code](#) / [Docs](#)

NERDSS-MPI | parallel particle-based reaction-diffusion simulation | [Code](#) / [Guide](#)

ioNERDSS | automated structure-to-simulation modeling | [Code](#) / [Docs](#)

EDUCATION

Ph.D. in Physics | Institute of Physics, Chinese Academy of Sciences

2019

B.S. in Physics | Nankai University

2014